Data Analysis for the Titanic Dataset with Pandas and Matplotlib

What is Matplotlib?

Its main job is to take your data (like numbers in a list or a Pandas `DataFrame`) and turn it into a **static picture** (a plot or chart).

It's incredibly powerful and customizable. Almost any plot you can imagine, you can build with Matplotlib.

The Most Important Concept: `fig` and `ax`

`fig, ax = plt.subplots()`. This is the "object-oriented" way to plot, and it's the most flexible.

Let's break down that one line: `fig, ax = plt.subplots(figsize=(10, 6))`

- **`plt.subplots()`:** This is a function that creates two things for you:
  1. **`fig` (The Figure):** This is the **entire window or page** your plot lives on. Think of it as the picture frame. You use `fig` to do things like save the plot to a file (e.g., `fig.savefig('my_plot.png')`).
  2. **`ax` (The Axes):** This is the **actual plot** itself—the part with the data, labels, and ticks. Think of it as the canvas inside the frame. This is what you'll use 99% of the time. You call methods on `ax` to create the plot, like `ax.bar()`, `ax.scatter()`, and `ax.set_title()`.

Analogy: `fig` is the picture frame, `ax` is the canvas you draw on.

Setup: Loading the Titanic Data

You can load the Titanic Dataset directly from `Kaggle` into the pandas `DataFrame`.

```
import matplotlib.pyplot as plt
import pandas as pd
import kaggle

# Load the dataset
kaggle.api.authenticate()
kaggle.api.dataset_download_files('yasserh/titanic-dataset', path='data/', unzip=True)
df = pd.read_csv('data/Titanic-Dataset.csv')

# Let's see what columns we have
print(df.head())
```

The Basic Plots

Here are the most common plots, what they're for, and how to make them with your `ax` object.

1. Bar Plot

- **What is it?** A plot that uses rectangular bars to compare values for different **categories**.
- **When to use it?** When you want to compare a numerical value (like a count or an average) across several distinct groups.

Titanic Example 01: Let's compare the **number of survivors in each passenger class (`pclass`).**

In [83]:
```
import pandas as pd
df = pd.read_csv('Titanic-Dataset.csv')
```

In [84]:
```
df
```

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Out[84]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows × 12 columns

In [85]:
```
df['Age'] = df['Age'].fillna(df['Age'].median())
```

In [86]:
```
import matplotlib.pyplot as plt
```

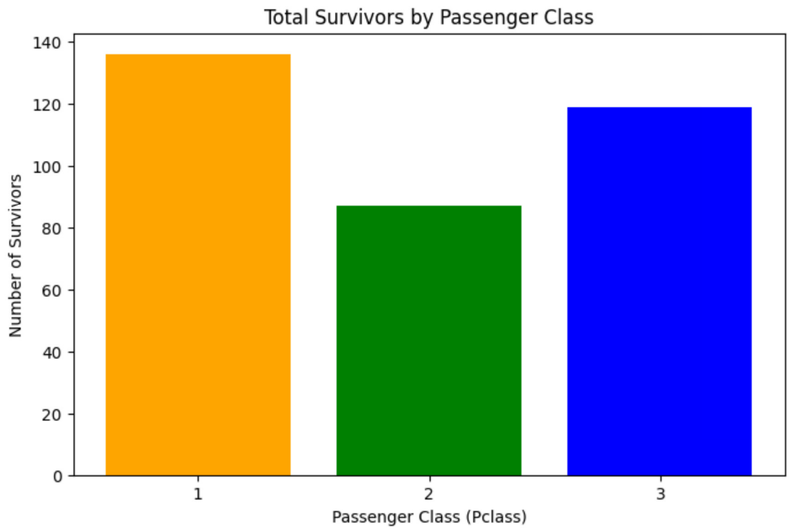
In [87]:
```
# --- setup ---
fig, ax = plt.subplots(figsize=(8, 5))

survivedByClass = df.groupby('Pclass')['Survived'].sum()


ax.bar(survivedByClass.index, survivedByClass.values, color = ['orange', 'green', 'blue'])

ax.set_title('Total Survivors by Passenger Class')
ax.set_xlabel('Passenger Class (Pclass)')
ax.set_ylabel('Number of Survivors')
ax.set_xticks([1, 2, 3]) # set clean ticks on the x-axis
```

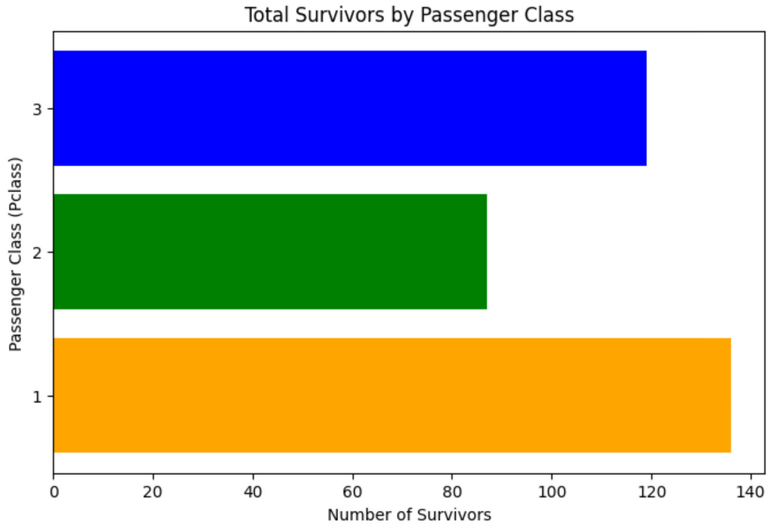
Out[87]:
```
[<matplotlib.axis.XTick at 0x1981c5d1590>,
 <matplotlib.axis.XTick at 0x1981c5d3c50>,
 <matplotlib.axis.XTick at 0x1981c628410>]
```



In [88]:
```
fig, ax = plt.subplots(figsize=(8, 5))
ax.barh(survivedByClass.index, survivedByClass.values, color = ['orange', 'green', 'blue'])

ax.set_title('Total Survivors by Passenger Class')
ax.set_xlabel('Number of Survivors')
ax.set_ylabel('Passenger Class (Pclass)')
ax.set_yticks([1, 2, 3]) # set clean ticks on the x-axis
```

Out[88]:
```
[<matplotlib.axis.YTick at 0x1981c64df90>,
 <matplotlib.axis.YTick at 0x1981c64fed0>,
 <matplotlib.axis.YTick at 0x1981c698690>]
```



Common Parameters for `ax.bar()`:

- `x`: The categories (our `pclass` index: `[1, 2, 3]`).
- `height`: The numerical value for each bar (our survivor counts).
- `color`: Sets the color of all bars (e.g., `color='skyblue'`).
- `edgecolor`: Sets the color of the bar outlines (e.g., `edgecolor='black'`).
- `label`: A name for this data, used if you call `ax.legend()`.

Pro-tip: Use `ax.barh()` for a **horizontal bar chart** (you'd use `y` and `width` parameters instead).

---

2. Histogram

- **What is it?** A plot that shows the **distribution** (or frequency) of a *single continuous* variable. It groups data into "bins" (buckets) and plots the count for each bin.
- **When to use it?** When you want to see the *shape* of your data. For example: "Are most passengers young? Are high fares common or rare?"

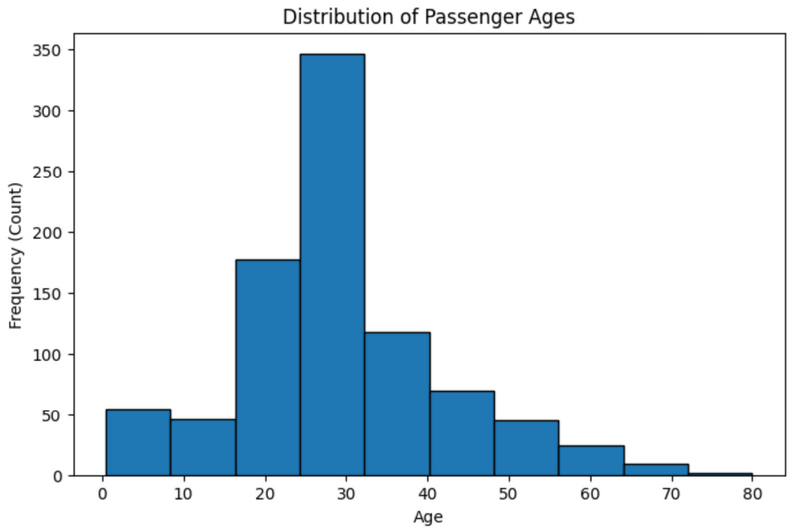
Titanic Example 02: Let's look at the **distribution of passenger ages.**

In [89]:
```
fig, ax = plt.subplots(figsize=(8, 5))

ax.hist(df['Age'], edgecolor = 'black')

ax.set_title('Distribution of Passenger Ages')
ax.set_xlabel('Age')
ax.set_ylabel('Frequency (Count)')
```

Out[89]:
```
Text(0, 0.5, 'Frequency (Count)')
```

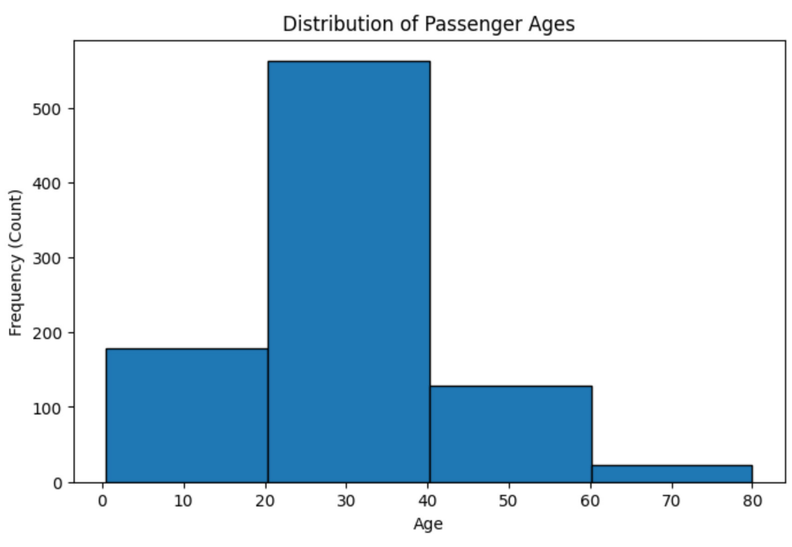


In [90]:
```
fig, ax = plt.subplots(figsize=(8, 5))

ax.hist(df['Age'], bins= 4, edgecolor = 'black')

ax.set_title('Distribution of Passenger Ages')
ax.set_xlabel('Age')
ax.set_ylabel('Frequency (Count)')
```

Out[90]: Text(0, 0.5, 'Frequency (Count)')



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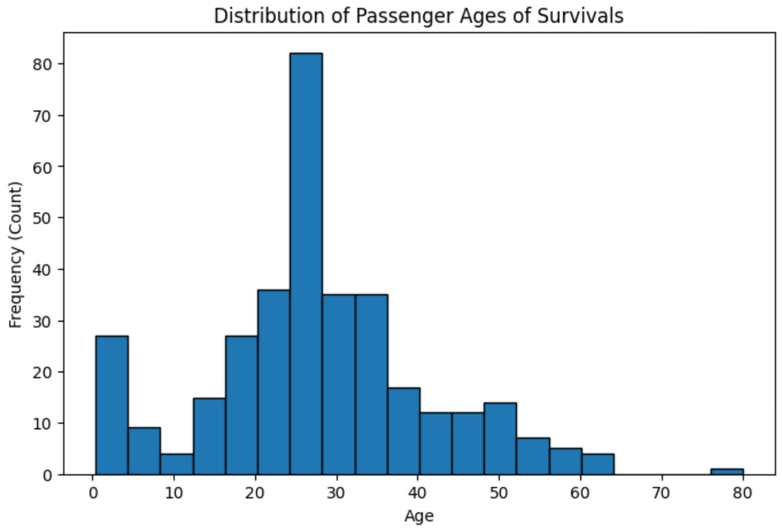
Titanic Example 03: Distribution of passenger ages of survivals.

In [91]:
```
fig, ax = plt.subplots(figsize=(8, 5))

ax.hist(df[df['Survived']==1]['Age'], bins= 20, edgecolor = 'black')

ax.set_title('Distribution of Passenger Ages of Survivals')
ax.set_xlabel('Age')
ax.set_ylabel('Frequency (Count)')
```

Out[91]: Text(0, 0.5, 'Frequency (Count)')



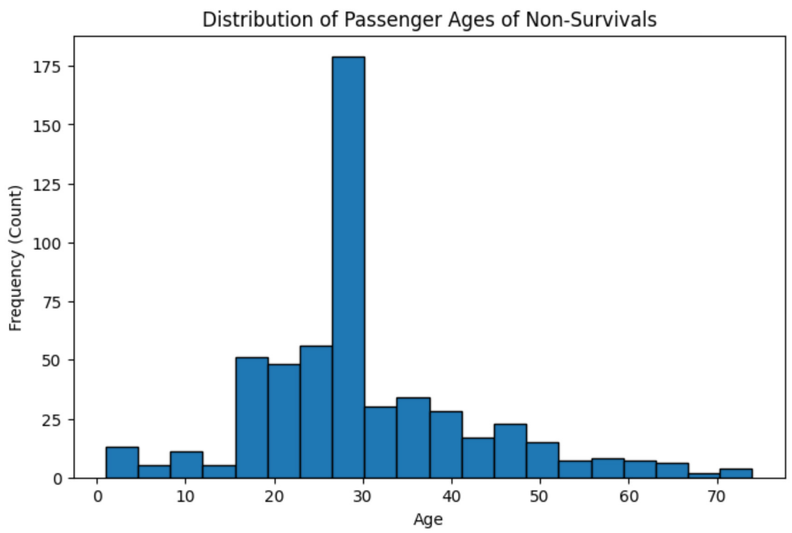
Titanic Example 04: Distribution of passenger ages of Non-Survivals.

In [92]:
```
fig, ax = plt.subplots(figsize=(8, 5))

ax.hist(df[df['Survived']==0]['Age'], bins= 20, edgecolor = 'black')

ax.set_title('Distribution of Passenger Ages of Non-Survivals')
ax.set_xlabel('Age')
ax.set_ylabel('Frequency (Count)')
```

Out[92]:
```
Text(0, 0.5, 'Frequency (Count)')
```



Titanic Example 05: Distribution of passenger ages of Survivals by Gender.

In [93]:
```
fig, ax = plt.subplots(1,2,figsize=(20, 5))

ax[0].hist(df[(df['Survived']==1) & (df['Sex']=='male')]['Age'], bins= 20, edgecolor = 'black')

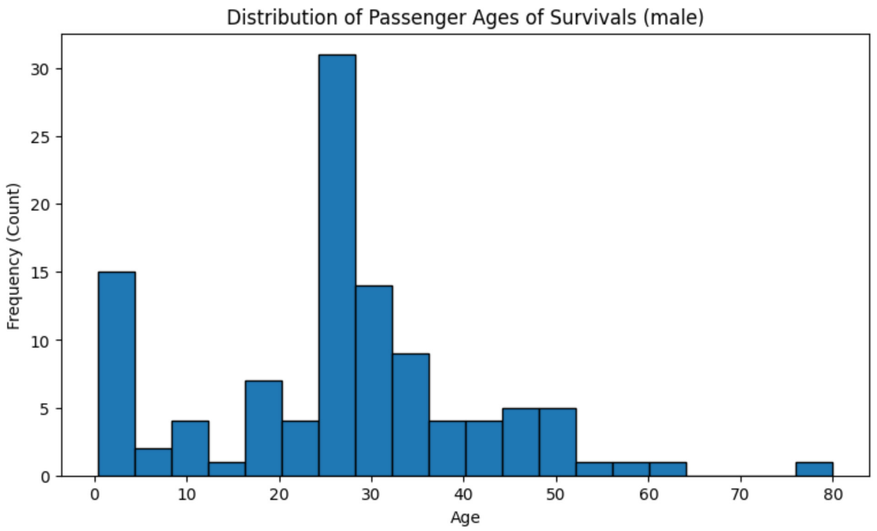
ax[0].set_title('Distribution of Passenger Ages of Survivals (male)')
ax[0].set_xlabel('Age')
ax[0].set_ylabel('Frequency (Count)')

ax[1].hist(df[(df['Survived']==1) & (df['Sex']=='female')]['Age'], bins= 20, edgecolor = 'black', color ='pink')

ax[1].set_title('Distribution of Passenger Ages of Survivals (female)')
ax[1].set_xlabel('Age')
ax[1].set_ylabel('Frequency (Count)')
```

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Out[93]: Text(0, 0.5, 'Frequency (Count)')





Common Parameters for `ax.hist()`:

- `x`: The (single) array or list of data to plot.
- `bins`: This is the most important one! It's the number of buckets to create.
  - `bins=20` (Create 20 equal-width bins).
  - `bins=[0, 10, 20, 30, 60, 80]` (Define the exact bin edges).
- `color`: The fill color of the bars.
- `edgecolor`: The outline color of the bars.
- `alpha`: Transparency (from `0.0` to `1.0`). Useful for overlapping histograms (e.g., `alpha=0.7`).

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3. Scatter Plot

- **What is it?** A plot that shows the value of individual data points by placing a dot for each one.
- **When to use it?** To see the **relationship between two continuous variables**.

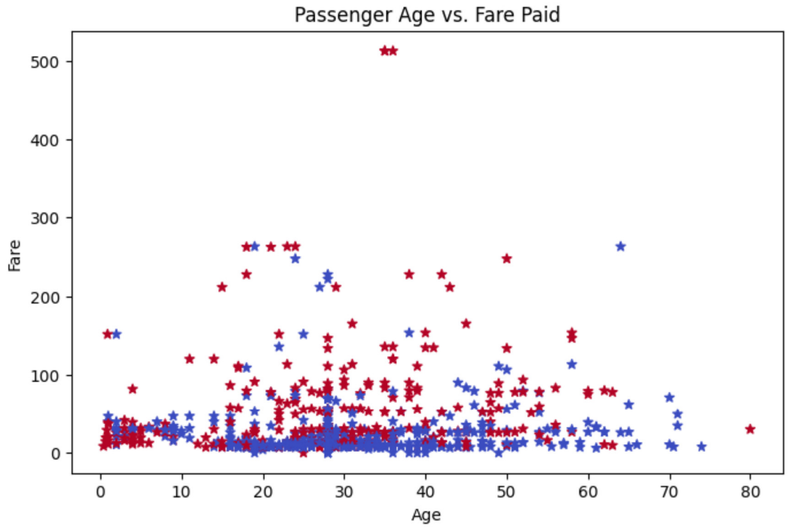
Titanic Example 06: Let's see if there's a relationship between a passenger's `age` and the `fare` they paid.

In [94]:
```
fig, ax = plt.subplots(figsize=(8, 5))

ax.scatter(x=df['Age'], y=df['Fare'], c=df['Survived'], marker='*', cmap='coolwarm')

ax.set_title('Passenger Age vs. Fare Paid')
ax.set_xlabel('Age')
ax.set_ylabel('Fare')
```

Out[94]: Text(0, 0.5, 'Fare')



Common Parameters for `ax.scatter()`:

- `x`, `y`: The two variables you want to compare.
- `s`: The **size** of each marker.
- `c`: The **color**. This is powerful! You can set it to a third variable.
  - *Example:* `c=df['Survived']` would color the dots differently based on whether the passenger survived.
- `cmap`: A colormap (like `'viridis'`) to use if `c` is a number (like `pclass`).
- `alpha`: Transparency (e.g., `alpha=0.5`), which is very useful when many dots overlap.
- `marker`: The shape of the marker (e.g., `marker='x'` or `marker='+'`).

---

4. Line Plot

- **What is it?** A plot that connects data points with a line.
- **When to use it?** Classically used to show a **trend over a continuous sequence**, most often **time**. The Titanic dataset doesn't have a great time variable, but it's perfect for things like stock prices over a year or temperature over a day.

Titanic Example 07: Since we don't have time, let's (a bit artificially) plot the **count of passengers by `age`**. *Note: A bar chart is probably better here, but this shows how `ax.plot()` works.*

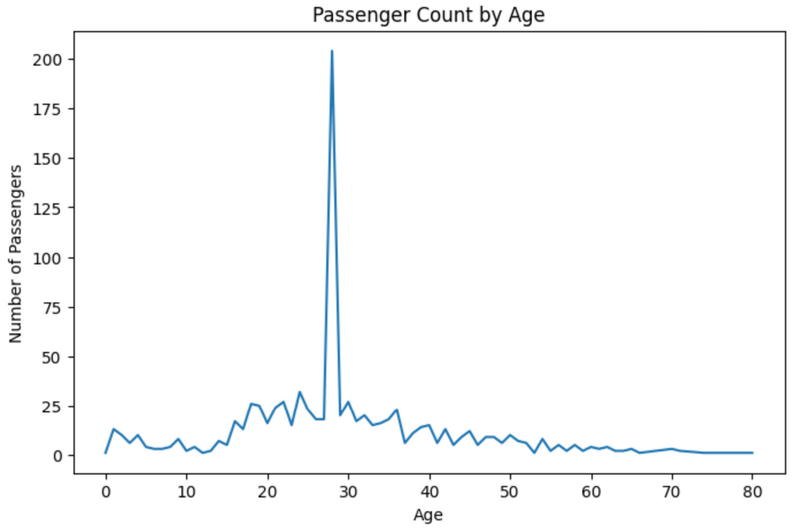
In [95]:
```
fig, ax = plt.subplots(figsize=(8, 5))

ageCounts = df['Age'].round().value_counts().sort_index()

ax.plot(ageCounts.index, ageCounts.values)

ax.set_title('Passenger Count by Age')
ax.set_xlabel('Age')
ax.set_ylabel('Number of Passengers')
```

Out[95]:
```
Text(0, 0.5, 'Number of Passengers')
```



Common Parameters for `ax.plot()`:

- `x`, `y`: The data points. `x` must be in order for the line to make sense.
- `color`: The color of the line (e.g., `color='red'`).
- `linestyle`: How the line is drawn:
  - `'-'` (solid, default)
  - `'--'` (dashed)
  - `':'` (dotted)
  - `'-.'` (dash-dot)
- `linewidth`: The thickness of the line (e.g., `linewidth=3`).
- `marker`: A marker for each data point (e.g., `marker='o'` for circles).
- `label`: A name for the legend (e.g., `label='Passengers'`). Call `ax.legend()` to show it.

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Matplotlib Cheat Sheet (for `ax` methods)

Plot Function	What It's For	Titanic Example
`ax.bar(x, height)`	Comparing a quantity across **categories**.	Number of survivors in each `pclass`.
`ax.hist(x)`	Distribution (frequency) of **one continuous** variable.	The `age` distribution of all passengers.
`ax.scatter(x, y)`	Relationship between **two continuous** variables.	`age` vs. `fare`.
`ax.plot(x, y)`	Trends over a **continuous (ordered)** sequence.	Stock price over time (or passenger count by *sorted* age).
`ax.boxplot(x)`	Distribution of a variable **across categories**.	`fare` distribution for each `pclass`.

Common Customization Commands (call these on `ax`):

- `ax.set_title("My Title")`: Sets the plot's main title.
- `ax.set_xlabel("X-Axis Label")`: Sets the label for the x-axis.
- `ax.set_ylabel("Y-Axis Label")`: Sets the label for the y-axis.
- `ax.legend()`: Displays the legend (requires `label=...` in your plot command).
- `ax.grid(True)`: Adds a background grid.
- `ax.set_xticklabels([...])`: Sets custom text labels for x-axis ticks.
- `ax.set_xticks([...])`: Sets the *location* of x-axis ticks.